

Difference between interpolation & numerical differentiation.

Interpolation:

- finding a function that connects specific points
 - simplest is linear
- math technique estimating unk data points between existing known points.

can only be by first

interpolating the function $y = y_1 + (x - x_1) \frac{(y_2 - y_1)}{(x_2 - x_1)}$

w/ a polynomial & then
differentiating the polynomial

Numerical Differentiation:

- process of approximating the derivative of a function using its values at discrete points
- estimate the rate of change of a function at a specific point, when the function is only known at discrete points
- methods include forward, backward, & central difference based on Taylor series expansions.